HIDDEN GROUNDWATER CONTAMINATION PATTERNS IN TAMIL NADU USING PYTHON

1. Dataset Information

Source: Central Ground Water Board (CGWB), Government of India / National Water Data Portal
Location: Tamil Nadu, India | Domain: Environmental / Water Quality Analytics
Observed period: 2020–2023 (to be reconfirmed on fresh load) | Previously inspected size: 433 rows, 47 original columns (including empty fields).

Data caution: The original proposal references a wider published time range and more chemical parameters than the inspected CSV supports. All final charts will use only verified available columns and units. If the rubric requires 500+ records, the current 433-record file does not meet that requirement.

2. Business / Real-World Problem

Groundwater monitoring data contains many chemical measurements collected across Tamil Nadu, but raw records do not clearly show which districts have comparatively high readings, how chemical parameters relate to each other, or where additional monitoring may be useful. Decision-makers need a simple, reliable summary to prioritize further testing and investigation.

3. Project Objectives

- Analyze groundwater quality across districts in Tamil Nadu.
- Compare districts to identify relatively high chemical measurements.
- Study relationships and changes in major groundwater chemical parameters.
- Identify unusual readings and recommend monitoring priorities.

4. Project Workflow

Stage	Planned work
Stage 1	Title, dataset details, objectives; import libraries once; load GitHub CSV once; initial EDA.
Stage 2	Check and remove confirmed duplicates once; inspect missing values; handle them without blanket row deletion; recheck; transform data; final EDA.
Stage 3	Central tendency + chart; variance/standard deviation + chart; skewness/kurtosis and optional outlier box plot; then 12 distinct visualizations, each in its own cell with interpretation.
Stage 4	3–5 evidence-based findings; descriptive, diagnostic, predictive (future scope), prescriptive analysis; recommendations; future enhancements.

5. Twelve Unique Main Visualizations

Type	Chart and purpose
Univariate (5)	1. pH histogram - acidity/alkalinity distribution 2. TDS histogram - dissolved-solids distribution 3. Uranium histogram - concentration distribution 4. District sample-count bar chart - monitoring coverage 5. pH-category count plot - acidic / neutral / alkaline sample counts

Bivariate (4)	6. District vs median TDS bar chart - district comparison 7. EC vs TDS scatter plot - chemical association 8. Sampling year vs median hardness line chart - observed time variation 9. District vs median uranium bar chart - district comparison
Multivariate (3)	10. Chemical-parameter correlation heatmap - several relationships 11. District × chemical-parameter standardized heatmap - district profiles 12. Year × district median TDS heatmap - variation across places and years

Note: The three mandatory statistical sections may include their own supporting charts; they will answer distinct statistical questions rather than repeat a main visualization. If a required chemical column lacks sufficient valid values, replace its chart with another verified parameter while preserving the chart count and purpose.

6. Key Findings / Insights to Establish from Actual Outputs

These are investigation targets, not completed findings. Exact district names, concentrations, trends, and percentages must be filled in after executing the notebook.

- District variation: Which districts have higher typical TDS, hardness, or uranium readings?
- Chemical relationships: Is EC strongly associated with TDS, and which other parameters correlate?
- Distribution and unusual readings: Which chemical parameters have skewed distributions or extreme observations?
- Time and coverage: How do available yearly measurements and district sample counts differ?
- Monitoring priorities: Which observed high readings merit repeat sampling or laboratory verification?

7. Practical Business Solutions / Decision Support

- Prioritize follow-up sampling at locations with consistently elevated verified measurements.
- Compare districts using medians and sample counts, not just isolated maximum values.
- Investigate unusual observations and confirm units, collection dates, and laboratory records before action.
- Improve monitoring coverage in underrepresented districts and maintain consistent sampling over time.
- Publish a concise periodic quality report or dashboard for monitoring teams.

8. Types of Analysis and Future Enhancement

Descriptive: Summarize samples, chemical values and district differences. Diagnostic: Explore observed relationships and anomalies without claiming causation. Predictive: Future extension only; no model is promised. Prescriptive: Suggest evidence-based follow-up monitoring.

Future enhancements: Obtain a larger/recent validated dataset, include more consistent station-level sampling, automate checks and reports, and consider predictive modelling if the data supports it.

9. Final Deliverable and Presentation Story

A reproducible Python analysis notebook with clear preprocessing, three statistical sections, 12 distinct visualizations, concise interpretations, and evidence-based findings and recommendations. Presentation flow: Problem → Objectives → Data cleaning → Statistics → District comparisons → Insights → Monitoring recommendations.

Scientific limitation: High readings or statistical associations alone do not prove contamination sources or establish drinking-water safety. Threshold comparisons require verified units and applicable standards.